

Workshop 4

Problem 1

Keywords: One-sided limits, Continuity, Limit Laws

Let

$$f(x) = \begin{cases} x^2 + ax + b & x < -1, \\ 2 & x = -1, \\ \frac{\tan(b(x+1))}{2x+2} & x > -1, \end{cases}$$

where a, b are some unspecified constants. For what values of a, b is f continuous at -1 ?

Problem 2

Keywords: Intermediate Value Theorem, Roots

Suppose $f(x) = x^3 + x - 1$.

- (a) Explain why f has a root in the interval $[0, 1]$.
- (b) Assume g is a continuous function on $[0, 1]$ such that $g(0) = 2$ and $g(1) = \frac{1}{2}$. Explain why the equation $f(x) = g(x)$ has a solution in the interval $[0, 1]$.
- (c) Suppose A is a constant and $h(x) = x^3 + x - 1 + Ax(x-1)(2x-1)$. Show that h has at least one root in the interval $[0, 1]$.
- (d) Calculate $h\left(\frac{1}{3}\right)$ and $h\left(\frac{2}{3}\right)$ in terms of A .
- (e) For what values of A do we have $h\left(\frac{1}{3}\right) > 0$ and $h\left(\frac{2}{3}\right) < 0$?
- (f) Show that if A is large enough, h must have three roots in the interval $[0, 1]$.

Problem 3.

Keywords: Horizontal Asymptotes, Limits at Infinity

Let $f(x) = \frac{3x+1}{x+\sqrt{4x^2+5}}$.

- (a) Compute $\lim_{x \rightarrow \infty} f(x)$.
- (b) Compute $\lim_{x \rightarrow -\infty} f(x)$. [Warning: $\sqrt{(-10)^2} = 10$, not -10 .]
- (c) How many different horizontal asymptotes can a function have? What about oblique asymptotes and vertical asymptotes?